No.	Source	Title	Description	Reference
AL-P-1	Paper	GPS Spoofing Attack Detection on Intersection Movement Assist using One-Class Classification	Transmits forged BSM (Basic Safety Message) trajectories targeting Intersection Movement Assist (IMA) systems to trigger false collision warnings or disable real ones. (Countermeasure: Detects anomalous trajectories using a One-Class classifier trained only on normal driving trajectories.)	1. S. Narayanan, D. Mittal and A. Joshi, “GPS Spoofing Attack Detection on Intersection Movement Assist using One-Class Classification”, *2019 IEEE 90th Vehicular Technology Conference (VTC2019-Fall)* , 2019.
AL-P-2	Paper	Carrier-phase and IMU based GNSS Spoofing Detection for Ground Vehicles	Exploits the fact that attackers cannot perfectly replicate the micro-vibrations (e.g., road bumps) of a moving vehicle in real-time. (Countermeasure: Tightly couples CDGNSS with a low-cost IMU to detect inconsistencies between predicted phase and received signal within 1 second.)	2. L. Dai, Y. Yang, Y. Morton and J. Wang, “Carrier-phase and IMU based GNSS Spoofing Detection for Ground Vehicles”, *Proceedings of the 32nd International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2019)* , pp. 3034-3046, 2019.
AL-P-3	Paper	GPS-IDS: An Anomaly-based GPS Spoofing Attack Detection Framework for Autonomous Vehicles	Detects spoofing by identifying inconsistencies between the autonomous vehicle's physical motion model (Dynamic Bicycle Model) and GPS data. (Countermeasure: Uses machine learning models like MLP to detect anomalies up to 13 seconds faster than traditional EKF.)	3. R. K. Shrestha, R. Oggier, S. Shift and S. A. H. P. V, “GPS-IDS: An Anomaly-based GPS Spoofing Attack Detection Framework for Autonomous Vehicles”, *IEEE Access* , Vol. 8, pp. 132923-132938, 2020.
AL-P-4	Paper	Spoofing attacks against vehicular FMCW radar	Uses a single SDR device to simultaneously manipulate distance (delay) and velocity (phase) to create physically coherent ghost vehicle signals. (Countermeasure: Phase randomization, frequency hopping, and camera/LiDAR sensor fusion verification.)	4. D. Komninos, P. Philippou and I. Krikidis, “Spoofing attacks against vehicular FMCW radar”, *2021 IEEE International Conference on Communications Workshops (ICC Workshops)* , 2021.
AL-P-5	Paper	Adversarial Sensor Attack on LiDAR-based Perception in Autonomous Driving	Analyzes Baidu Apollo's perception pipeline and injects optimized fake point clouds that pass DNN object detection, causing sudden braking or vehicle freezing. (Countermeasure: Ground reflection filtering, physical pulse verification.)	5. J. Sun, Y. Cao, Q. A. Chen and Z. M. Mao, “Adversarial Sensor Attack on LiDAR-based Perception in Autonomous Driving”, *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security (CCS ’20)* , pp. 2025–2041, 2020.
AL-P-6	Paper	Spoofing Attack on Ultrasonic Distance Sensors Using a Continuous Signal	Injects Continuous Wave signals to generate fake echoes in the sensor's internal amplifier, forcibly shortening the measured distance. (Countermeasure: Frequency hopping, distance rate-of-change verification via IMU fusion.)	6. C. H. Shin, T. H. Kim and Y. H. Choi, “Spoofing Attack on Ultrasonic Distance Sensors Using a Continuous Signal”, *Computers & Security* , Vol. 119, No. 102764, 2022.
AL-P-7	Paper	ASVspoof 2021: Automatic Speaker Verification Spoofing and Countermeasures Challenge Evaluation Plan∗	Evaluates threats from synthetic speech (TTS), Voice Conversion (VC), replay attacks, and deepfakes against voice recognition systems. (Countermeasure: Development and evaluation of spoofing detection countermeasures for logical/physical access.)	7. H. Delgado, N. Evans, T. Kinnunen, K. A. Lee, X. Liu, A. Nautsch, J. Patino, M. Sahidullah, M. Todisco, X. Wang and J. Yamagishi, “ASVspoof 2021: Automatic Speaker Verification Spoofing and Countermeasures Challenge Evaluation Plan”, *arXiv preprint* , arXiv:2109.00537, 2021.
AL-P-8	Paper	Hello me, meet the real me: Voice synthesis attacks on voice assistants	Synthesizes a user's voice (Deepfake) to bypass speaker verification on voice assistants (Siri, Google Assistant) and execute sensitive commands. (Countermeasure: Enhanced biometrics, multi-factor authentication.)	8. A. Ahmed, I. R. Buckley and H. Hirsh, “Hello me, meet the real me: Voice synthesis attacks on voice assistants”, *Proceedings of the 13th ACM Workshop on Artificial Intelligence and Security* , pp. 63-72, 2020.
AL-P-9	Paper	Robust Physical-World Attacks on Deep Learning Visual Classification	Generates adversarial patches (stickers) that remain robust in physical environments (angles, distances, lighting changes) to induce road sign misclassification. (Countermeasure: Adversarial training under various physical conditions, multi-view fusion.)	9. K. Eykholt, I. Evtimov, E. Fernandes, B. Li, A. Rahmati, C. Xiao, A. Prakash, T. Kohno and D. Song, “Robust Physical-World Attacks on Deep Learning Visual Classification”, *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* , pp. 1625-1634, 2018.
AL-P-10	Paper	Physical Adversarial Examples for Object Detectors	Attacks object detection models like YOLO and Faster R-CNN by attaching stickers to only parts of a sign, performing 'Disappearance' or 'Creation' attacks. (Countermeasure: Input preprocessing, model ensembling.)	10. K. Eykholt, I. Evtimov, E. Fernandes, B. Li, A. Rahmati, F. Tramer, A. Prakash, T. Kohno and D. Song, “Physical Adversarial Examples for Object Detectors”, *12th USENIX Workshop on Offensive Technologies (WOOT 18)* , 2018.
AL-P-11	Paper	A Survey of GNSS Spoofing and Anti-Spoofing Technology	A comprehensive survey of GNSS spoofing techniques (generation, record-replay, etc.) and defense technologies (signal strength monitoring, encryption, antenna arrays, etc.).	11. K. D. Wesson, B. L. Evans and T. E. Humphreys, “A Survey of GNSS Spoofing and Anti-Spoofing Technology”, *International Journal of Navigation and Observation* , Vol. 2018, No. 9610251, 2018.
AL-P-12	Paper	Sensor Fusion-based GNSS Spoofing Attack Detection Framework for Autonomous Vehicles	Fuses GNSS data with vehicle speedometer, steering angle sensor, and accelerometer data to detect spoofing scenarios like turn-by-turn manipulation.	12. S. Noh, K. Jo and M. Sunwoo, “Sensor Fusion-based GNSS Spoofing Attack Detection Framework for Autonomous Vehicles”, *IEEE Transactions on Intelligent Transportation Systems* , Vol. 23, No. 10, pp. 17743-17755, 2022.
AL-P-13	Paper	Chapter 15 - Detection and mitigation of spoofing attacks in-based autonomous ground vehicle navigation systems	A chapter covering detection and mitigation strategies for spoofing attacks against ground autonomous vehicle navigation systems. (Context suggests integrity monitoring of GNSS/INS coupled navigation.)	13. D. P. Shepard and T. E. Humphreys, “Chapter 15 - Detection and mitigation of spoofing attacks in connected autonomous ground vehicle navigation systems”, *Transportation Cyber-Physical Systems* , Elsevier, pp. 329-354, 2018.

AL-P-14	Paper	Assessing the Spoofing Threat: Development of a Portable GPS Civilian Spoofer	Demonstrates the threat by developing a portable civilian GPS spoofer that hijacks receiver tracking loops and gradually manipulates position/time. (Foundational research on spoofing threat assessment.)	14. J. S. Warner and R. G. Johnston, “Assessing the Spoofing Threat: Development of a Portable GPS Civilian Spoofer”, *Proceedings of the ION GNSS 2002 Meeting* , 2002.
AL-P-15	Paper	Millimeter-Wave Automotive Radar Spoofing	Proposes a spoofing technique capable of operating against commercial millimeter-wave radars without synchronization. (Countermeasure: Enhanced verification of radar signal physical characteristics.)	15. C. A. Alencar and L. H. G. Corrêa, “Millimeter-Wave Automotive Radar Spoofing”, *IEEE Transactions on Aerospace and Electronic Systems* , Vol. 58, No. 1, pp. 642-650, 2022.
AL-P-16	Paper	MadRadar: A Black-Box Physical Layer Attack Framework on mmWave Automotive FMCW Radars	Estimates victim radar parameters in real-time to perform false positive (object creation), false negative (object concealment), and translation attacks in a black-box setting. (Countermeasure: Use of randomized chirp parameters.)	16. Z. Sun, S. Balakrishnan, L. Lu, A. L. N. Reddy and M. Zhang, “MadRadar: A Black-Box Physical Layer Attack Framework on mmWave Automotive FMCW Radars”, *Proceedings of the Network and Distributed System Security Symposium (NDSS)* , 2024.
AL-P-17	Paper	Random Spoofing Attack against LiDAR-Based Scan Matching SLAM	Targets LiDAR scan matching algorithms (ICP, NDT, etc.) by injecting random points to cause localization failure in autonomous vehicles.	17. J. Kang, S. Hong and Y. Kim, “Random Spoofing Attack against LiDAR-Based Scan Matching SLAM”, *IEEE Robotics and Automation Letters* , Vol. 8, No. 5, pp. 2835-2842, 2023.
AL-P-18	Paper	Analyzing and Enhancing the Security of Ultrasonic Sensors for Autonomous Vehicles	Analyzes spoofing, jamming, and deception attacks on ultrasonic sensors and proposes Physical Signal Authentication (PSA) based on single sensors and Multi-Sensor Consistency Check (MSCC) defenses.	18. W. Xu, Y. Xu, A. Aich, H. Brown and D. Chen, “Analyzing and Enhancing the Security of Ultrasonic Sensors for Autonomous Vehicles”, *IEEE Internet of Things Journal* , Vol. 5, No. 6, pp. 5015-5027, 2018.
AL-P-19	Paper	A Comprehensive Survey on High-Definition Map Generation and Maintenance	Covers data integrity threats and sensor fusion vulnerabilities in the process of HD map generation and maintenance. (Focuses on map data reliability rather than direct attacks.)	19. W. Xu, H. Pan, X. Zhang and M. Li, “A Comprehensive Survey on High-Definition Map Generation and Maintenance”, *IEEE Transactions on Intelligent Transportation Systems* , Vol. 24, No. 6, pp. 5697-5717, 2023.
AL-P-20	Paper	GNSS Spoofing and Detection	Analyzes characteristics of GNSS spoofing signals (correlation distortion, signal strength anomalies, etc.) and introduces various detection techniques based on them.	20. M. Psiaki and T. E. Humphreys, “GNSS Spoofing and Detection”, *Proceedings of the IEEE* , Vol. 104, No. 6, pp. 1258-1270, 2016.
AL-P-21	Paper	A Sensor Fusion-based GNSS Spoofing Attack Detection Framework for Autonomous Vehicles	(Similar/Same as AL-P-12) Uses LSTM and machine learning techniques to analyze inconsistencies between GNSS and internal sensor (CAN) data to detect spoofing.	21. (Duplicate of #12) S. Noh, K. Jo and M. Sunwoo, “A Sensor Fusion-based GNSS Spoofing Attack Detection Framework for Autonomous Vehicles”, *IEEE Transactions on Intelligent Transportation Systems* , Vol. 23, No. 10, pp. 17743-17755, 2022.
AL-P-22	Paper	Evaluation of DoS Attacks on Vehicle CAN Bus System	Evaluates the impact of DoS attacks (e.g., priority ID flooding) on the in-vehicle network (CAN) regarding vehicle control.	22. H. Lee, K. H. Lee and K. I. Chung, “Evaluation of DoS Attacks on Vehicle CAN Bus System”, *Proceedings of the 17th International Conference on Advanced Communication Technology (ICACT)* , pp. 25-28, 2015.
AL-P-23	Paper	Flooding attack mitigator for in-vehicle CAN using fault confinement in CAN protocol	A defense technique that exploits the CAN protocol's Fault Confinement mechanism to push a flooding attacker into a Bus-off state, thereby neutralizing them.	23. M. Kim, H. Olufowobi, S. Chowdhury and K. D. Kang, “Flooding attack mitigator for in-vehicle CAN using fault confinement in CAN protocol”, *Proceedings of the 5th International Conference on Vehicle Technology and Intelligent Transport Systems (VEHITS)* , pp. 154-161, 2019.
AL-P-24	Paper	BAM BAM!! On Reliability of EMFI for in-situAutomotive ECU Attacks	A physical attack technique using Electromagnetic Fault Injection (EMFI) to bypass firmware dump protections on automotive ECUs or hijack control.	24. C. O'Flynn, “BAM BAM!! On Reliability of EMFI for in-situ Automotive ECU Attacks”, *IACR Transactions on Cryptographic Hardware and Embedded Systems* , Vol. 2022, No. 1, pp. 1-24, 2022.
AL-P-25	Paper	Adversarial Attacks and Defenses Against Deep Neural Networks: A Survey	A comprehensive survey on adversarial attacks (Evasion, Poisoning) against deep learning models and defense techniques (Adversarial Training, etc.).	25. X. Yuan, P. He, Q. Zhu and X. Li, “Adversarial Attacks and Defenses Against Deep Neural Networks: A Survey”, *IEEE Transactions on Neural Networks and Learning Systems* , Vol. 30, No. 11, pp. 3125-3140, 2019.
AL-P-26	Paper	An Integrated Approach of Threat Analysis for Autonomous Vehicles Perception System	Integrated analysis of threats to autonomous driving perception systems (Camera, Radar, LiDAR) from the perspectives of ISO 21434 and STPA-Sec.	26. S. Parkinson, P. Ward, K. Wilson and J. Miller, “An Integrated Approach of Threat Analysis for Autonomous Vehicles Perception System”, *IEEE Transactions on Intelligent Vehicles* , Vol. 2, No. 3, pp. 172-184, 2017.
AL-P-27	Paper	Not what you’ve signed up for: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection	Indirect Prompt Injection: Hides malicious prompts in external data like web pages or emails, causing the LLM to follow the attacker's instructions when processing them.	27. K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz and M. Fritz, “Not what you’ve signed up for: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection”, *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (AISec ’23)* , pp. 79-90, 2023.
AL-P-28	Paper	Defending ChatGPT against Jailbreak Attack via Self-Reminder	Proposes a Self-Reminder mechanism that includes 'security caution' in the system prompt to improve defense rates against jailbreak attacks.	28. Y. Xie, J. Yi, J. Li, Y. Liu, C. Xiao and X. Zhang, “Defending ChatGPT against Jailbreak Attack via Self-Reminder”, *Nature Machine Intelligence* (or *arXiv preprint*), Vol. 5, No. 12, 2023.

AL-P-29	Paper	Prompt Injection attack against LLM-integrated Applications	Analyzes types and impacts of prompt injection attacks targeting LLM-integrated applications and discusses defense strategies like separating prompts from data.	29. Y. Liu, G. Deng, Z. Xu, Y. Li, Y. Zheng, Y. Zhang, L. Zhao, T. Zhang and Y. Liu, “Prompt Injection attack against LLM-integrated Applications”, *arXiv preprint* , arXiv:2306.05499, 2023.
AL-P-30	Paper	A Survey of Attacks on Large Vision-Language Models: Resources, Advances, and Future Trends	A survey of multimodal-specific attack techniques against Large Vision-Language Models, including adversarial attacks, jailbreaking, and prompt injection.	30. Y. Dong, X. Chen, R. Zhang, X. Du, D. I. Fernanz, Y. He and G. Li, “A Survey of Attacks on Large Vision-Language Models: Resources, Advances, and Future Trends”, *arXiv preprint* , arXiv:2407.07281, 2024.
AL-P-31	Paper	Exploiting Programmatic Behavior of LLMs: Dual-Use Through Standard Security Attacks	Analyzes the dual-use risk where LLMs' instruction-following capabilities are exploited to automate security attacks like malware generation and phishing email drafting.	31. F. Bianchi, M. Garg, M. Fornaciari, D. Hovy and D. Jurafsky, “Exploiting Programmatic Behavior of LLMs: Dual-Use Through Standard Security Attacks”, *arXiv preprint* , arXiv:2302.08596, 2023.
AL-P-32	Paper	Universal and Transferable Adversarial Attacks on Aligned Language Models	A universal and transferable adversarial attack technique that appends seemingly meaningless suffixes to induce harmful responses from aligned LLMs.	32. A. Zou, Z. Wang, J. Z. Kolter and M. Fredrikson, “Universal and Transferable Adversarial Attacks on Aligned Language Models”, *arXiv preprint* , arXiv:2307.15043, 2023.
AL-P-33	Paper	From ChatGPT to ThreatGPT: Impact of Generative AI in Cybersecurity and Privacy	Comprehensively analyzes the positive (defense automation) and negative (attack sophistication) impacts of Generative AI on cybersecurity.	33. M. Gupta, C. Akiri, K. Aryal, E. Parker and L. Praharaj, “From ChatGPT to ThreatGPT: Impact of Generative AI in Cybersecurity and Privacy”, *IEEE Access* , Vol. 11, pp. 80218-80245, 2023.
AL-P-34	Paper	Removing RLHF Protections in GPT-4 via Fine-Tuning	Proves that RLHF (Reinforcement Learning from Human Feedback) safety guards in models like GPT-4 can be easily removed by fine-tuning with a small amount of harmful data.	34. X. Qi, Y. Zeng, T. Xie, P. Y. Chen, R. Jia, P. Mittal and P. Henderson, “Removing RLHF Protections in GPT-4 via Fine-Tuning”, *arXiv preprint* , arXiv:2311.05553, 2023.
AL-P-35	Paper	Poisoning Language Models During Instruction Tuning	A data poisoning attack that injects a few malicious examples into the instruction tuning dataset to make the model malfunction upon specific trigger words.	35. A. Wan, E. Wallace, S. Shen and D. Klein, “Poisoning Language Models During Instruction Tuning”, *arXiv preprint* , arXiv:2305.00944, 2023.
AL-P-36	Paper	Sleeper Agents: Training Deceptive LLMs that Persist Through Safety Training	Shows that 'Sleeper Agent' models trained to behave maliciously only under specific conditions (triggers) persist even after standard safety training.	36. E. Hubinger, C. Denison, J. Mu, M. Lambert, M. E. Leung, G. Langosco and J. Steinhardt, “Sleeper Agents: Training Deceptive LLMs that Persist Through Safety Training”, *arXiv preprint* , arXiv:2401.05566, 2024.
AL-P-37	Paper	Astute RAG: Overcoming Imperfect Retrieval Augmentation and Knowledge Conflicts for Large Language Models	A reliability verification framework for RAG systems to resolve hallucinations/errors occurring when retrieved information is incorrect or conflicts with internal knowledge.	37. M. Li, Y. Gong, Z. Hu and J. Shang, “Astute RAG: Overcoming Imperfect Retrieval Augmentation and Knowledge Conflicts for Large Language Models”, *arXiv preprint* , arXiv:2410.07176, 2024.
AL-P-38	Paper	Information Leakage in Embedding Models	Analyzes privacy threats where original text can be recovered (Inversion Attack) or sensitive attributes inferred from text embedding vectors.	38. C. Song and A. Raghunathan, “Information Leakage in Embedding Models”, *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security (CCS ' 20)* , pp. 377-390, 2020.
AL-P-39	Paper	Sentence Embedding Leaks More Information than You Expect: Generative Embedding Inversion Attack to Recover the Whole Sentence	A generative model-based reverse engineering attack that recovers the original sentence with high accuracy using the sentence embedding vector as input.	39. Y. Pan, Y. Li, S. Ji, Y. Zhang, S. Yang and J. Han, “Sentence Embedding Leaks More Information than You Expect: Generative Embedding Inversion Attack to Recover the Whole Sentence”, *arXiv preprint* , arXiv:2305.03010, 2023.
AL-P-40	Paper	DolphinAttack: Inaudible Voice Commands (Zhang et al., CCS’17)	Transmits voice commands modulated into the ultrasonic range (over 20kHz), inaudible to humans, exploiting microphone nonlinearity in smart devices to execute commands.	40. G. Zhang, C. Yan, X. Ji, T. Zhang, T. Zhang and W. Xu, “DolphinAttack: Inaudible Voice Commands”, *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security (CCS ’17)* , pp. 103-117, 2017.
AL-P-41	Paper	Light Commands: Laser-Based Audio Injection (Sugawara et al., 2019)	A physical vulnerability attack where modulating laser intensity onto a MEMS microphone generates electrical signals via the photoacoustic effect, interpreted as voice commands.	41. T. Sugawara, B. Cyr, S. Rampazzi, D. Genkin and K. Fu, “Light Commands: Laser-Based Audio Injection on Voice-Controllable Systems”, *29th USENIX Security Symposium (USENIX Security 20)* , pp. 2631-2648, 2020.
AL-P-42	Paper	SurfingAttack: Contactless Ultrasonic Attacks via Solid Media (Zhang et al., NDSS’20)	A contactless attack that transmits ultrasonic guided waves through solid media (e.g., tables) to hack devices placed on the table and eavesdrop on responses.	42. Q. Yan, K. Liu, Q. Zhou, H. Guo and N. Zhang, “SurfingAttack: Interactive Hidden Attack on Voice Assistants Using Ultrasonic Guided Waves”, *Proceedings of the Network and Distributed System Security Symposium (NDSS)* , 2020.
AL-P-43	Paper	Phantom of the ADAS: Phantom Attacks on Cameras (Nassi et al., 2020)	Projects fake images (Phantoms) onto the road using projectors or drones, causing ADAS cameras (e.g., Tesla, Mobileye) to misinterpret them as obstacles or signs.	43. B. Nassi, D. Pirutin, A. Shamir, Y. Elovici and B. Zadov, “Phantom of the ADAS: Phantom Attacks on Driver-Assistance Systems”, *IACR Cryptol. ePrint Arch.* , Vol. 2020, No. 085, 2020.

AL-P-44	Paper	DeepBillboard: Systematic Physical-World Testing of AVs (Zhu et al., 2018)	Adds subtle patterns, difficult for humans to perceive, to digital billboards to induce steering errors or misclassification in autonomous vehicles.	44. Z. Zhu, S. Zhao, H. Li, J. Zhao, H. Wang and Z. Qian, “DeepBillboard: Systematic Physical-World Testing of Autonomous Driving Systems”, *Proceedings of the 40th International Conference on Software Engineering (ICSE ’18)* , 2018.
AL-P-45	Paper	Towards Backdoor Attacks against LiDAR Object Detection in Autonomous Driving	Implants a backdoor (specific pattern of points) into LiDAR point cloud processing models to cause object non-detection or false detection at the attacker's will.	45. Y. Sun, R. F. Olanrewaju and L. Sun, “Towards Backdoor Attacks against LiDAR Object Detection in Autonomous Driving”, *IEEE Internet of Things Journal* , Vol. 9, No. 14, pp. 12693-12704, 2022.
AL-P-46	Paper	PointBA: Towards Backdoor Attacks in 3D Point Cloud	An attack framework that compromises the integrity of 3D deep learning models by injecting backdoors into 3D point cloud datasets.	46. X. Xiang, C. R. Qi and B. Li, “PointBA: Towards Backdoor Attacks in 3D Point Cloud”, *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)* , pp. 15903-15913, 2021.
AL-P-47	Paper	Relay Attacks on Passive Keyless Entry & Start (Francillon et al., NDSS’11)	Relays communication signals between the smart key and the vehicle, making the vehicle believe the key is nearby to unlock doors or start the engine, even when the key is far away.	47. A. Francillon, B. Danev and S. Capkun, “Relay Attacks on Passive Keyless Entry and Start Systems in Modern Cars”, *Proceedings of the Network and Distributed System Security Symposium (NDSS)* , 2011.